

Sustainable City Planner

Name: _____

Date: _____

Year 6 Maths — Number — Mathematical Modelling

ACARA v9.0: AC9M6N09

Learning Intention

I am learning to use mathematical modelling to solve a practical problem, formulating and solving the problem and justifying my choices.

Success Criteria

- I can use area calculations to plan a sustainable space.
- I can formulate and solve a practical modelling problem.
- I can justify my design choices using maths.

Plan, Calculate, Justify

Mathematical modelling means: understand the problem, choose the maths needed (like area), solve it step by step, then explain and justify your solution.

Worked Example

A 100m^2 park needs a rectangular garden using 20% of the space.

$20\% \text{ of } 100\text{m}^2 = 20\text{m}^2$.

A garden $5\text{m} \times 4\text{m} = 20\text{m}^2$ fits perfectly.

Part A — Warm Up

What is 25% of 200m^2 ? _____

True or False: Mathematical modelling includes justifying your solution, not just calculating an answer. _____

Calculate the area of a $10\text{m} \times 6\text{m}$ park space. _____

Part B — Core Skills

A council reserves 30% of a 500m^2 block for solar panels. Calculate the area reserved.

A rectangular community garden is 15m long and needs an area of 90m^2 . Find its width. _____

A bike path uses 12% of a $1,200\text{m}^2$ park. Calculate the bike path's area. _____

Explain why a city planner might choose a rectangular shape for a new park instead of an irregular shape.

Word Bank

sustainable • model • justify • area • percentage • formulate • solve • practical problem

Working Space — Design a Simple Rectangular Park (Include Length, Width and Area)

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Part C — Challenge: Designing Geelong's New Eco-Park

The Geelong City Council has a rectangular block of land, 40m long and 25m wide, for a new sustainable eco-park. They want 40% of the space for native trees, 25% for a community vegetable garden, and the rest for open grass space.

Q8. Calculate the total area of the block, then calculate the area reserved for native trees.

Q9. Calculate the area reserved for the community vegetable garden.

Q10. Calculate the remaining area for open grass space, and justify whether this design seems like a reasonable, balanced use of the land for a sustainable park.

Answer Key — Part A

A1: 50m^2 (25% of $200 = 50$).

A2: True — mathematical modelling includes justifying and communicating the solution, not just the calculation.

A3: 60m^2 (10×6).

Answer Key — Part B

B1: 150m^2 (30% of $500 = 150$).

B2: 6m ($90 \div 15 = 6$).

B3: 144m^2 (12% of $1,200 = 144$).

B4: A rectangular shape is easier to measure and calculate area for, and often makes efficient use of space with straight paths and clear zones, compared to an irregular shape which is harder to plan and measure accurately.

Answer Key — Part C

C8: Total area = $40 \times 25 = 1,000\text{m}^2$. Native trees = 40% of $1,000 = 400\text{m}^2$.

C9: 250m^2 (25% of $1,000 = 250$).

C10: Remaining grass space = $1,000 - 400 - 250 = 350\text{m}^2$ (35% of the park). This seems like a reasonably balanced design, giving nearly equal priority to nature (trees), community use (garden) and open space (grass), supporting a genuinely sustainable, multi-purpose park.

I Can Checklist

☐ I can use area to plan a sustainable space.

☐ I can justify my design choices with maths.

☐ I can formulate and solve a modelling problem.